

Global Superstore Sales Dashboard

Power BI Project Documentation

Abstract

This project presents an interactive Power BI dashboard developed using the Global Superstore dataset. The dashboard analyses sales, profit, customer behaviour, product performance, category trends, and regional performance from 2011 to 2014. The objective is to provide business stakeholders with actionable insights for data-driven decision-making through KPI monitoring and interactive reporting.

Varshitha Rachapudi
Varshitharachapudi20052gmail.com

Overview:

The Global Superstore Sales Dashboard is an interactive Business Intelligence solution developed using Microsoft Power BI. The dashboard analyses retail sales data from 2011 to 2014 and provides a comprehensive view of business performance across sales, profit, customers, products, categories, regions, and customer segments.

The project was designed to transform raw transactional data into meaningful insights through KPI tracking, trend analysis, and interactive visualizations. The dashboard enables stakeholders to monitor business performance, identify top-performing products and customers, evaluate regional performance, and make data-driven decisions.

The solution consists of two dashboard pages:

Executive Overview:

Focuses on high-level business metrics such as Total Sales, Total Profit, Total Orders, Average Order Value (AOV), Profit Margin, sales trends, category performance, and regional performance.

Customer & Product Analysis:

Focuses on customer behavior, product performance, and segment analysis through Top-N customer and product insights.

This project demonstrates practical skills in Power BI, DAX, data visualization, KPI design, business analysis, and dashboard development.

Business Problem

A retail company operating across multiple regions wants to analyze its sales performance, profitability, customer behaviour, and product performance. The objective is to identify top-performing categories, regions, customers, and products to support data-driven business decisions.

The dashboard helps stakeholders understand:

- Overall business performance
- Sales and profit trends
- Category and regional performance
- Customer purchasing behaviour
- Product profitability
- Segment-wise contribution

Dataset Overview

The Global Superstore dataset contains retail transactional data from 2011 to 2014. The dataset includes information related to orders, customers, products, sales, profit, categories, regions, and customer segments.

Dataset Features:

- Order ID
- Order Date
- Customer Name
- Segment
- Region
- Category
- Product Name
- Sales
- Profit
- Discount

Tools Used:

- Power BI
- DAX (Data Analysis Expressions)
- Data Visualization
- Data Modeling

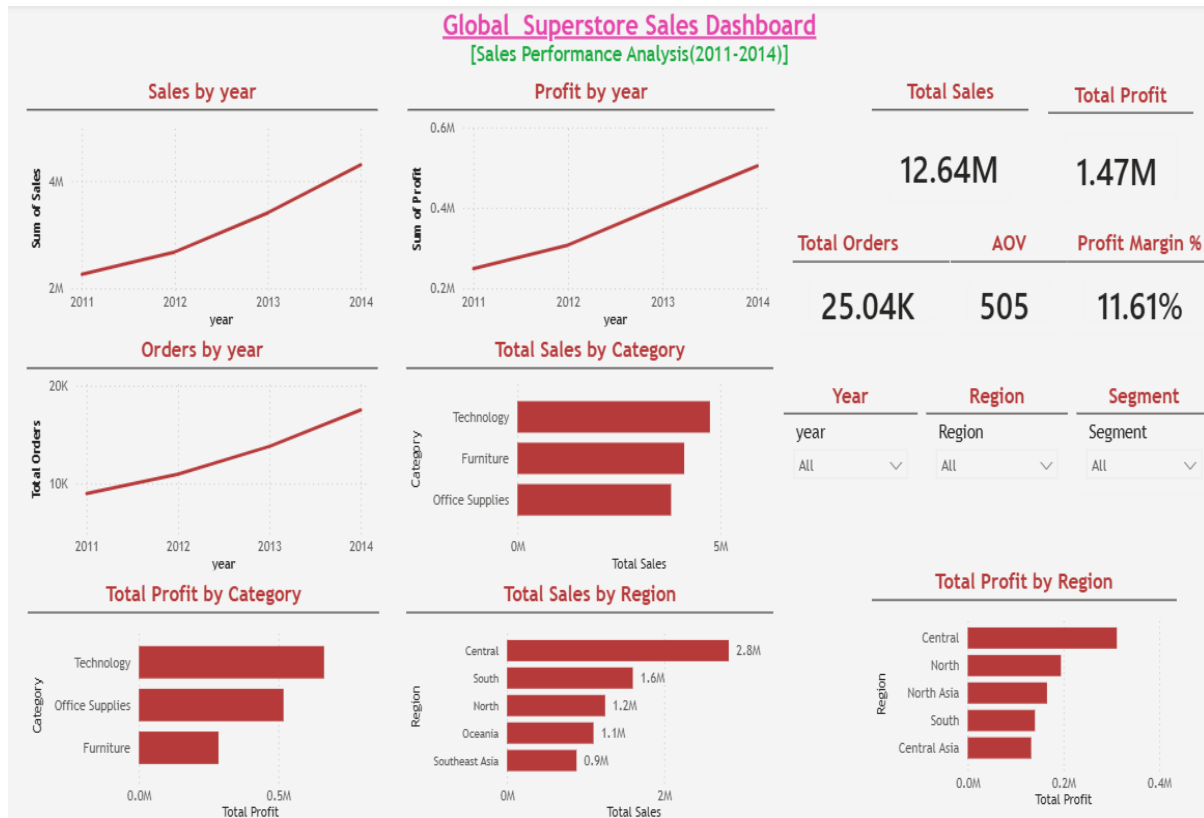
KPI Summary:

KPI	VALUE
Total Sales	12.64M
Total Profit	1.47M
Total Orders	25.04K
Average Order Value (AOV)	505
Profit Margin	11.61%

KPI Definitions:

- Total Sales: Total revenue generated from all orders.
- Total Profit: Total profit earned from sales.
- Total Orders: Total number of unique customer orders.
- Average Order Value (AOV): Average revenue generated per order.
- Profit Margin: Percentage of revenue converted into profit.

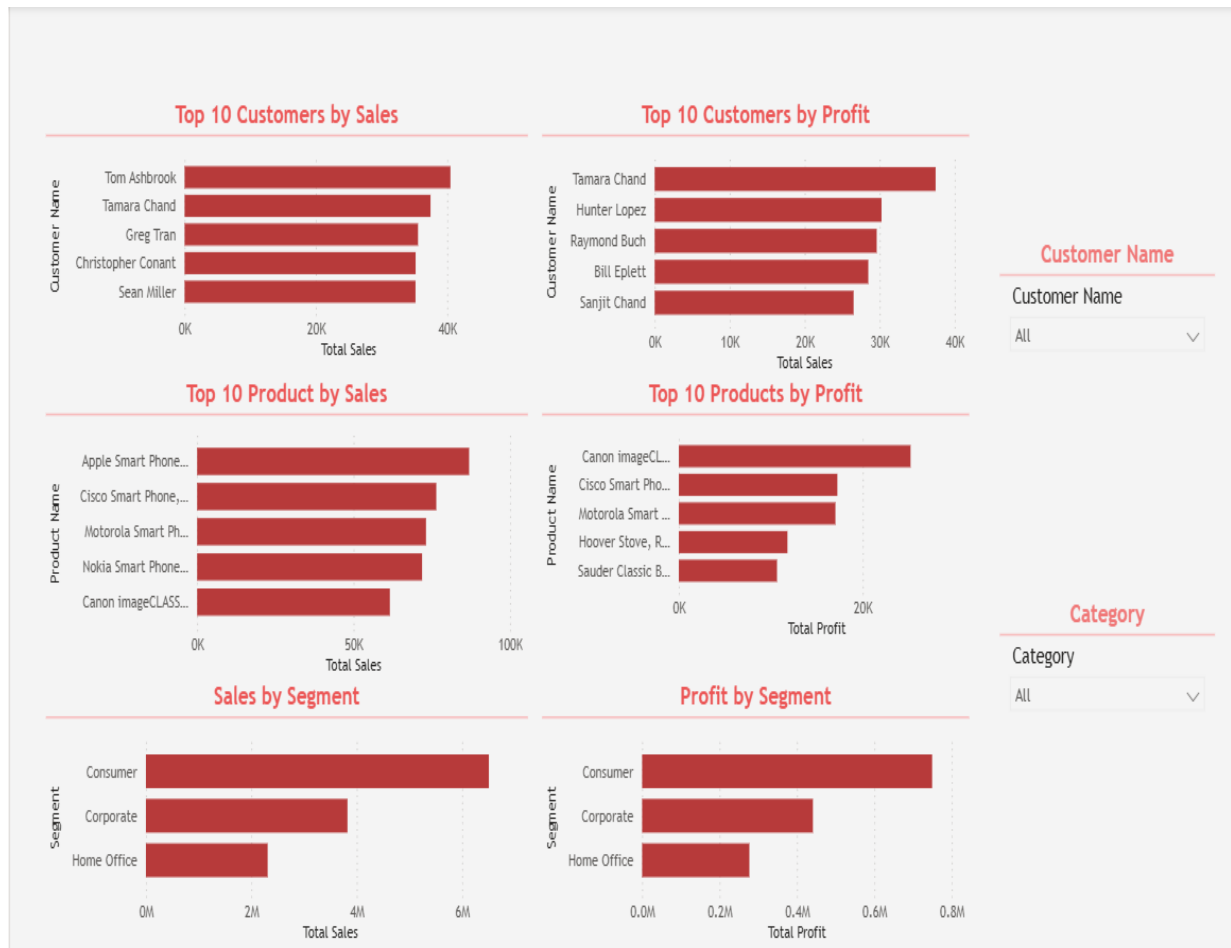
Executive Overview Dashboard:



Dashboard Components:

- KPI Cards
- Sales Trend Analysis
- Profit Trend Analysis
- Orders Trend Analysis
- Sales by Category
- Profit by Category
- Sales by Region
- Profit by Region
- Interactive Slicers

Customer & Product Analysis Dashboard:



Dashboard Components:

- Top 10 Customers by Sales
- Top 10 Customers by Profit
- Top 10 Products by Sales
- Top 10 Products by Profit
- Sales by Segment
- Profit by Segment
- Customer Filter
- Category Filter

Business Insights:

Sales Insights

- Sales increased consistently from 2011 to 2014.
- Total Sales reached 12.64M.
- Order volume increased every year, indicating business growth.

Profit Insights

- Total Profit reached 1.47M.
- Profit showed a positive growth trend throughout the period.
- The overall Profit Margin was 11.61%.

Category Insights

- Technology generated the highest sales.
- Technology generated the highest profit.
- Furniture generated lower profitability compared to Technology.

Regional Insights

- Central region generated the highest sales.
- Central region generated the highest profit.
- Regional performance varied significantly across locations.

Customer Insights

- Tom Ashbrook generated the highest sales.
- Tamara Chand generated the highest profit.
- High-sales customers were not always the most profitable customers.

Product Insights

- Apple Smart Phone was among the top-selling products.
- Canon imageCLASS generated the highest profit.
- High sales did not always result in high profit.

Segment Insights

- Consumer segment generated the highest sales.
- Consumer segment generated the highest profit.
- Home Office contributed the least sales and profit.

Challenges Faced

- Creating DAX measures for KPI calculations.
- Designing a two-page dashboard without overcrowding visuals.
- Implementing Top-N customer and product analysis.
- Creating interactive slicers and filters for dashboard navigation.
- Transforming raw transactional data into meaningful business insights.

Skills Demonstrated:

Technical Skills:

- Power BI Dashboard Development
- DAX Measure Creation
- KPI Design and Analysis
- Data Modeling
- Interactive Slicer Implementation
- Top-N Analysis
- Business Intelligence Reporting

Analytical Skills

- Trend Analysis
- Customer Analysis
- Product Analysis
- Category Analysis
- Regional Analysis
- Segment Analysis
- Business Insight Generation

DAX Measures Created:

Measure	Purpose
Total Sales	Calculates total revenue
Total Profit	Calculates total profit
Total Orders	Counts unique orders
AOV	Calculates average order value
Profit Margin %	Calculates business profitability percentage

Conclusion:

The dashboard successfully transformed transactional sales data into actionable business insights. Through KPI monitoring, trend analysis, customer segmentation, and product performance evaluation, the project demonstrates the practical application of Power BI and DAX for business intelligence reporting.